

Reverse Engineer Any APK File

First of all, install Reverse Engineering tools

Download Diva app from github

<https://github.com/0xArab/diva-apk-file/blob/main/DivaApplication.apk>

```
# apt install apktool zipalign openjdk-11-jdk
```

```
# apktool d DivaApplication.apk
```

```
(root@kali)-[/home/kali/Downloads]
# apktool d DivaApplication.apk
I: Using Apktool 2.7.0-dirty on DivaApplication.apk
I: Loading resource table ...
I: Decoding AndroidManifest.xml with resources ...
I: Loading resource table from file: /root/.local/share/apktool/framework/1.apk
I: Regular manifest package ...
I: Decoding file-resources ...
I: Decoding values */* XMLs ...
I: Baksmaling classes.dex ...
I: Copying assets and libs ...
I: Copying unknown files ...
I: Copying original files ...
```

After decompile you can check folder data

If you want you can make few changes in file

```
(root@kali)-[/home/kali/Downloads]
# ls
DivaApplication  DivaApplication.apk

(root@kali)-[/home/kali/Downloads]
# ls DivaApplication
AndroidManifest.xml  apktool.yml  lib  original  res  smali
```

Now rebuild apk

```
# apktool b DivaApplication
```

If you got error then try this

```
# apktool b DivaApplication --use-aapt2
```

```
(root@kali)-[/home/kali/Downloads]
# apktool b DivaApplication --use-aapt2
I: Using Apktool 2.7.0-dirty
I: Checking whether sources has changed ...
I: Smaling smali folder into classes.dex ...
I: Checking whether resources has changed ...
I: Building resources ...
W: aapt: brut.common.BrutException: brut.common.BrutException: Could
  not find java.nio.channels.spi.AbstractSelectableChannel
  (defaulting to $PATH binary)
I: Copying libs ... (/lib)
I: Building apk file ...
I: Copying unknown files/dir ...
I: Built apk into: DivaApplication/dist/DivaApplication.apk
```



The apk file is created in dist folder

```
(root@kali)-[/home/kali/Downloads]
# ls DivaApplication/dist
DivaApplication.apk
```

Demo apk rebuild successful, this app is not signed, so it will not execute anywhere

Now let's create signature file and add owner details

```
# keytool -genkey -v -keystore ctf.keystore -alias ctfKeystore -keyalg RSA -keysize
2048 -validity 10000
```

```
(root@kali)-[/home/kali/test]
# keytool -genkey -v -keystore ctf.keystore -alias ctfKeystore -keyalg RSA -keysize 2048 -validity 10000
Enter keystore password:
Re-enter new password:
Enter the distinguished name. Provide a single dot (.) to leave a sub-component empty or press ENTER to use the
.
What is your first and last name?
[Unknown]:
What is the name of your organizational unit?
[Unknown]:
What is the name of your organization?
[Unknown]:
What is the name of your City or Locality?
[Unknown]:
What is the name of your State or Province?
[Unknown]:
What is the two-letter country code for this unit?
[Unknown]:
Is CN=Unknown, OU=Unknown, O=Unknown, L=Unknown, ST=Unknown, C=Unknown correct?
[no]: yes

Generating 2,048 bit RSA key pair and self-signed certificate (SHA384withRSA) with a validity of 10,000 days
for: CN=Unknown, OU=Unknown, O=Unknown, L=Unknown, ST=Unknown, C=Unknown
[Storing ctf.keystore]
```

(ctf.keystore is signature file)

```
(root@kali)-[/home/kali/Downloads]
# ls
ctf.keystore  DivaApplication  DivaApplication.apk
```

Now let's assign this signature file to apk file

```
# jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore ctf.keystore
DivaApplication/dist/DivaApplication.apk ctfKeystore
```

```
(root@kali)-[/home/kali/Downloads]
# jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore ctf.keystore DivaApplication/
dist/DivaApplication.apk ctfKeystore
Enter Passphrase for keystore:
adding: META-INF/MANIFEST.MF
adding: META-INF/CTFKEYST.SF
adding: META-INF/CTFKEYST.RSA
```

```
# jarsigner -verify -verbose -certs DivaApplication/dist/DivaApplication.apk
```

```
(root@kali)-[/home/kali/Downloads]
# jarsigner -verify -verbose -certs DivaApplication/dist/DivaApplication.apk

39894 Wed Nov 05 12:09:18 EST 2025 META-INF/MANIFEST.MF
40015 Wed Nov 05 12:09:18 EST 2025 META-INF/CTFKEYST.SF
1266 Wed Nov 05 12:09:18 EST 2025 META-INF/CTFKEYST.RSA
m ? 13460 Wed Nov 05 12:02:48 EST 2025 lib/armeabi-v7a/libdivajni.so
m ? 5488 Wed Nov 05 12:02:48 EST 2025 lib/mips/libdivajni.so
m ? 5164 Wed Nov 05 12:02:48 EST 2025 lib/x86/libdivajni.so
m ? 13452 Wed Nov 05 12:02:48 EST 2025 lib/armeabi/libdivajni.so
m ? 6048 Wed Nov 05 12:02:48 EST 2025 lib/mips64/libdivajni.so
m ? 5512 Wed Nov 05 12:02:48 EST 2025 lib/arm64-v8a/libdivajni.so
```

```
zipalign -v 4 DivaApplication/dist/DivaApplication.apk DivaApplication-aligned.apk
```

```
(root@kali)-[/home/kali/Downloads]
# zipalign -v 4 DivaApplication/dist/DivaApplication.apk DivaApplication-aligned.apk
Verifying alignment of DivaApplication-aligned.apk (4) ...
50 META-INF/MANIFEST.MF (OK - compressed)
11878 META-INF/CTFKEYST.SF (OK - compressed)
23804 META-INF/CTFKEYST.RSA (OK - compressed)
24940 lib/armeabi-v7a/libdivajni.so (OK - compressed)
30479 lib/mips/libdivajni.so (OK - compressed)
32021 lib/x86/libdivajni.so (OK - compressed)
33554 lib/armeabi/libdivajni.so (OK - compressed)
```

```
(root@kali)-[/home/kali/Downloads]
# ls
ctf.keystore  DivaApplication  DivaApplication-aligned.apk  DivaApplication.apk
(root@kali)-[/home/kali/Downloads]
```

Now install this app and test